The Quiz Hack Sheet

Brownian Motion, Martingales & Stochastic Integrals

Ordered by frequency on the quiz format: the top hacks solve most questions.

Notation: $B = (B_t)$ standard BM, $(\mathcal{F}_t)$ its filtration, $s \wedge t = \min(s,t)$, $\Delta = B_t - B_s$.

Tier 1 — The workhorses (expect 2–4 questions each)

Hack 1: Itô isometry — kill the square

$$\mathbb{E}\left[\left(\int_0^t H_s \,\mathrm{d}B_s\right)^2\right] = \int_0^t \mathbb{E}[H_s^2]\,\mathrm{d}s$$

Any "$\mathbb{E}[(\int \cdots \mathrm{d}B)^2]$" question is this: square the integrand, take expectation, integrate $\mathrm{d}s$. Stochastic integrals also have mean zero, so this is the variance too.

Example. $\mathbb{E}[(\int_0^t e^s\,\mathrm{d}B_s)^2] = \int_0^t e^{2s}\,\mathrm{d}s = \frac{e^{2t}-1}{2}$.

Hack 2: The moment table (memorize cold)

For $B_s \sim N(0,s)$:

$\mathbb{E}[B_s] = \mathbb{E}[B_s^3] = 0$ (all odd)	$\mathbb{E}[B_s^2] = s$	$\mathbb{E}[B_s^4] = 3s^2$	$\mathbb{E}[B_s^6] = 15s^3$
$\mathbb{E}[e^{\theta B_s}] = e^{\theta^2 s/2}$	$\mathbb{E}[B_s B_t] = s \wedge t$	$\mathbb{E}[Z^{2n}] = (2n-1)!! = 1, 3, 15, \ldots$	

Example. Inside an isometry: $\mathbb{E}[(1+B_s)^2] = 1 + 2\cdot 0 + s = 1 + s$; $\mathbb{E}[(2+B_s)^2] = 4 + s$.

Hack 3: Absorb B_t into the integral

$$B_t = \int_0^t 1\,\mathrm{d}B_s, \qquad cB_t = \int_0^t c\,\mathrm{d}B_s$$

When a question mixes B_t with a stochastic integral, merge into ONE integral, then apply Hack 1.

Example. $\mathbb{E}[(B_t + \int_0^t B_s\,\mathrm{d}B_s)^2] = \mathbb{E}[(\int_0^t (1+B_s)\,\mathrm{d}B_s)^2] = \int_0^t (1+s)\,\mathrm{d}s = t + \frac{t^2}{2}$.

Hack 4: Exponential martingale — match the drift

$e^{\theta B_t - \frac{\theta^2}{2}t}$ is a martingale. For $X_t = B_t + \mu t$: $e^{\theta X_t}$ martingale $\iff \theta = -2\mu$.

Example. $X_t = B_t + t$ ($\mu = 1$): answer e^{-2X_t}. $X_t = B_t - 2t$: $\theta = 4$, i.e. e^{4X_t}. Zero computation.

Trap. $\mu > 0$ and θ wrong $\Rightarrow$ sub/supermartingale: check $\mathbb{E}[e^{\theta B_t + \theta\mu t}] = e^{(\theta\mu + \theta^2/2)t}$; increasing = sub, decreasing = super.

Hack 5: The martingale dictionary

Process	Martingale version
B_t^2	$B_t^2 - t$
B_t^3	$B_t^3 - 3tB_t$
$e^{\theta B_t}$	$e^{\theta B_t - \theta^2 t/2}$
M_t^2 (M Gaussian mart.)	$M_t^2 - \Gamma(t,t)$
$B_t^{(1)} B_t^{(2)}$ (indep.)	already a martingale
$W_t B_t^{(2)}$ (corr ρ)	$W_t B_t^{(2)} - \rho t$
$\int_0^t H_s\,\mathrm{d}B_s$	already a martingale
$\int_0^t B_s\,\mathrm{d}s$	never (finite variation, non-constant)

Mean shortcut: compute $m(t) = \mathbb{E}[X_t]$. Increasing $\Rightarrow$ not a martingale; decreasing $\Rightarrow$ supermartingale; constant $\Rightarrow$ martingale candidate — on multiple choice this usually decides it.

Example. Is e^{B_t} a martingale? $\mathbb{E}[e^{B_t}] = e^{t/2}$ increasing $\Rightarrow$ no, submartingale. One line.

Hack 6: Scaling — reduce to B_1

$$B_t \overset{d}{=} \sqrt{t}\, B_1$$

Turns any marginal probability/moment at time t into a question about one $N(0,1)$.
Example. $P(B_t \le \sqrt{2}) = P(B_1 \le \sqrt{2/t}) \to P(B_1 \le 0) = \frac{1}{2}$ as $t \to \infty$. $\mathbb{E}[B_s^4] = s^2 \mathbb{E}[B_1^4] = 3s^2$.
Limit trichotomy for $\lim_t P(B_t \le c\, t^\alpha)$: $\alpha < \frac{1}{2} \Rightarrow \frac{1}{2}$; $\alpha = \frac{1}{2} \Rightarrow \Phi(c)$; $\alpha > \frac{1}{2}, c > 0 \Rightarrow 1$.

Tier 2 — Mixed moments & structure questions

Hack 7: Add-and-subtract (independent increments)

Write $B_t = \Delta + B_s$ with $\Delta = B_t - B_s \sim N(0, t-s)$ independent of $\mathcal{F}_s$; expand; products factorize; odd powers of Δ die.
Example. $\mathbb{E}[B_s^2 B_t^2] = \mathbb{E}[B_s^2(\Delta + B_s)^2] = \underbrace{\mathbb{E}[B_s^2]\mathbb{E}[\Delta^2]}_{s(t-s)} + \underbrace{\mathbb{E}[B_s^4]}_{3s^2} + 2\underbrace{\mathbb{E}[B_s^3]}_{0}\mathbb{E}[\Delta] = s(t-s) + 3s^2$.

Hack 8: Tower property — condition on the latest time

Peel from the outside in with

$$\mathbb{E}[B_t \mid \mathcal{F}_s] = B_s, \quad \mathbb{E}[B_t^2 \mid \mathcal{F}_s] = B_s^2 + (t-s), \quad \mathbb{E}[B_t^3 \mid \mathcal{F}_s] = B_s^3 + 3(t-s)B_s.$$

Example. $\mathbb{E}[B_s B_t B_u] = \mathbb{E}[B_s B_t\, \mathbb{E}[B_u \mid \mathcal{F}_t]] = \mathbb{E}[B_s B_t^2] = \mathbb{E}[B_s(B_s^2 + t - s)] = 0$.

Hack 9: Symmetry kills odd totals

$B \overset{d}{=} -B$ as a process, so for a monomial $B_{t_1}^{a_1} \cdots B_{t_k}^{a_k}$: $a_1 + \cdots + a_k$ odd $\Rightarrow \mathbb{E}[\cdot] = 0$. Also $\mathbb{E}[\sin B_t] = 0$, $\mathbb{E}[B_t e^{-B_t^2}] = 0$, etc.
Example. $\mathbb{E}[B_1^2 B_2^3]$: total power 5 odd $\Rightarrow 0$. But $\mathbb{E}[B_1^3 B_2^5]$: total 8 even $\Rightarrow$ must compute (Hacks 7–8).

Hack 10: Independence splitting

If X is $\mathcal{F}_s$-measurable and g depends only on increments after s: $\mathbb{E}[X \cdot g] = \mathbb{E}[X]\, \mathbb{E}[g]$.
Example. $\mathbb{E}[B_s^2\, e^{B_t - B_s}] = \mathbb{E}[B_s^2]\, \mathbb{E}[e^{B_t - B_s}] = s\, e^{(t-s)/2}$.

Hack 11: Quadratic variation lookup

$\left\langle \int_0^\cdot H_s \, \mathrm{d}B_s \right\rangle_t = \int_0^t H_s^2 \, \mathrm{d}s$, $\langle B \rangle_t = t$, $\langle \sigma B \rangle_t = \sigma^2 t$, $\langle \text{finite variation} \rangle_t = 0$.
Square the integrand, swap $\mathrm{d}B_s \to \mathrm{d}s$. Drift never changes it: $\langle B_\cdot + \mu \cdot \rangle_t = t$. Elimination: $\langle X \rangle$ is increasing & finite variation — any option containing $\mathrm{d}B_s$ is wrong.
Example. $X_t = \int_0^t B_s \, \mathrm{d}B_s \Rightarrow \langle X \rangle_t = \int_0^t B_s^2 \, \mathrm{d}s$. Done.

Hack 12: Time-changed BM $X_t = f(t) B_{g(t)}$

Always centred Gaussian (g increasing, f, g deterministic), with $\mathrm{cov}(X_s, X_t) = f(s)f(t)\, g(s \wedge t) = f(s)f(t)g(s)$ $(s < t)$. Elimination: kill every option with nonzero mean first.
Example. $e^{-t} B_{e^{2t}}$: $\mathrm{cov} = e^{-s} e^{-t} e^{2s} = e^{-(t-s)}$ — stationary OU-type covariance.

Hack 13: Bridge-type covariance: four min's

For $X_t = B_t + \alpha t B_1$, $s \le t \le 1$: expand into four terms, $\mathbb{E}[B_u B_v] = u \wedge v$ each:

$$\mathbb{E}[X_s X_t] = s + \alpha t s + \alpha s t + \alpha^2 s t.$$

Example. $\alpha = -1$ (bridge): $s - ts - st + st = s(1-t)$. $\alpha = +1$: $s(1+3t)$. Track the signs.

Trap. A bridge is pinned: variance $t(1-t)$ vanishes at $t=0$ AND $t=1$. Any option not dying at the endpoints is wrong.

Hack 14: Two-valued stopping times

For $T = a\,\mathbf{1}_A + b\,\mathbf{1}_{A^c}$, $a<b$, $A = \{\text{event about } B_u\}$: T stopping time $\iff A \in \mathcal{F}_a \iff u \le a$.
Example. $2\cdot\mathbf{1}_{\{B_2>1\}} + 4\cdot\mathbf{1}_{\{B_2\le 1\}}$: event at 2, smaller value 2, $2\le 2$ ✓. $1\cdot\mathbf{1}_{\{B_2>1\}} + \dots$ at time 1 you can't know B_2 ✗.
Smell test: hitting times of levels/closed sets ✓; argmax, last time, "never again", future-peeking ✗.

Hack 15: Indicators: probability + symmetry

$\mathbb{E}[\mathbf{1}_A] = P(A)$, $\mathbf{1}_A^2 = \mathbf{1}_A$, $P(B_t\le 0) = P(B_t \ge 0) = \frac{1}{2}$.
Example. $\mathbb{E}[(\int_0^1 \mathbf{1}_{\{B_t\ge 0\}}\,\mathrm{d}B_t)^2] \overset{H1}{=} \int_0^1 \mathbb{E}[\mathbf{1}^2]\,\mathrm{d}t = \int_0^1 P(B_t\ge 0)\,\mathrm{d}t = \frac{1}{2}$. Three hacks, three lines.

Hack 16: Nested integrals — isometry twice, inside out

For $\mathbb{E}[(\int_0^t X_s\,\mathrm{d}B_s)^2]$ with X itself a stochastic integral: inner isometry first for $\mathbb{E}[X_s^2]$, feed into the outer.
Example. $X_t = \int_0^t e^s\,\mathrm{d}B_s$: inner $\mathbb{E}[X_s^2] = \frac{e^{2s}-1}{2}$; outer $\int_0^t \frac{e^{2s}-1}{2}\,\mathrm{d}s = \frac{1}{4}(e^{2t}-1) - \frac{t}{2}$.

Tier 3 — Integration by parts & the $t^3/3$ family

Hack 17: Integration by parts — $\int_0^t B_s\,\mathrm{d}s$ as a Wiener integral

The classical $\int u\,\mathrm{d}v = [uv] - \int v\,\mathrm{d}u$ with $u=s$, $v=B_s$ (valid — one factor is smooth, so no Itô correction):

$$tB_t = \int_0^t B_s\,\mathrm{d}s + \int_0^t s\,\mathrm{d}B_s \quad\Longrightarrow\quad \boxed{\int_0^t B_s\,\mathrm{d}s = \int_0^t (t-s)\,\mathrm{d}B_s}$$

Now integrated BM is an integral of a *deterministic* function against $\mathrm{d}B$: Gaussian, centred, and every second-moment question becomes a one-line isometry.
Example. $\mathrm{var}\big(\int_0^t B_s\,\mathrm{d}s\big) = \int_0^t (t-s)^2\,\mathrm{d}s = \frac{t^3}{3}$.
Example. Intuition: a kick $\mathrm{d}B_s$ at time s stays inside the running integral for the remaining $t-s$ — that is the kernel.

Hack 18: The other key identity $\int_0^t B_s\,\mathrm{d}B_s = \frac{B_t^2 - t}{2}$

From $\mathrm{d}(B_t^2) = 2B_t\,\mathrm{d}B_t + \mathrm{d}t$. Converts between the two most common quiz objects instantly, and re-explains why $B_t^2 - t$ is a martingale (it is a stochastic integral).
Example. $\mathbb{E}[(\int_0^t B_s\,\mathrm{d}B_s)^2]$ two ways: isometry $\int_0^t s\,\mathrm{d}s = \frac{t^2}{2}$; or $\frac{1}{4}\mathbb{E}[(B_t^2-t)^2] = \frac{1}{4}(3t^2 - 2t^2 + t^2) = \frac{t^2}{2}$. Two hacks agreeing = answer confirmed.

Hack 19: Itô's formula — closed forms for $\int f'(B)\,\mathrm{d}B$

For smooth f (the template — memorize):

$$f(B_t) = f(0) + \int_0^t f'(B_s)\,\mathrm{d}B_s + \tfrac{1}{2}\int_0^t f''(B_s)\,\mathrm{d}s$$

Recipe: differentiate twice, plug in, rearrange for whichever integral the question asks about. The naive calculus answer minus a $\frac{1}{2}f''$ correction — the correction is $(\mathrm{d}B)^2 = \mathrm{d}t$ leaking in.

f	f'	$\frac{1}{2}f''$	Itô gives	Solved for the integral
x^2	$2x$	1	$B_t^2 = 2\int B\,\mathrm{d}B + t$	$\int_0^t B_s\,\mathrm{d}B_s = \frac{B_t^2 - t}{2}$
x^3	$3x^2$	$3x$	$B_t^3 = 3\int B^2\,\mathrm{d}B + 3\int B\,\mathrm{d}s$	$\int_0^t B_s^2\,\mathrm{d}B_s = \frac{B_t^3}{3} - \int_0^t B_s\,\mathrm{d}s$
x^4	$4x^3$	$6x^2$	$B_t^4 = 4\int B^3\,\mathrm{d}B + 6\int B^2\,\mathrm{d}s$	$\int_0^t B_s^3\,\mathrm{d}B_s = \frac{B_t^4}{4} - \frac{3}{2}\int_0^t B_s^2\,\mathrm{d}s$

Example. ($f = x^3$): $f' = 3x^2, f'' = 6x, f(0) = 0 \Rightarrow B_t^3 = 3\int_0^t B_s^2\,\mathrm{d}B_s + 3\int_0^t B_s\,\mathrm{d}s$; divide by 3, rearrange.
Mean check: LHS $\int f'(B)\,\mathrm{d}B$ has mean 0. Row 3: $\mathbb{E}[B_t^4] = 3t^2$ and $\frac{3}{2}\int_0^t s\,\mathrm{d}s = \frac{3t^2}{4}$, so $\frac{3t^2}{4} - \frac{3t^2}{4} = 0$ ✓.
Trap. $B_t^3 - 3\int_0^t B_s\,\mathrm{d}s$ and $B_t^3 - 3tB_t$ are BOTH martingales — reconciled by Hack 17. Their difference $3\int_0^t s\,\mathrm{d}B_s$ is itself a martingale.

Master Forms — the general Itô toolkit (parent of Hacks 4, 5, 17, 18, 19, 22, 33)

Convention. Write $f = f(x,t)$ with the *first slot* x *= space, second slot* t *= time.* Along the path we substitute the process into the first slot, so *every* derivative below is evaluated at the point (X_s, s) (or (B_s, s) for BM). Smoothness $C^{2,1}$: twice in x, once in t.

M0 — What $\mathrm{d}\langle X\rangle_t$ is, and the Itô multiplication table. $\langle X\rangle_t$ is the *quadratic variation*: the accumulated variance from the Brownian part only. Compute it by squaring the differential and applying

$$\mathrm{d}t\cdot\mathrm{d}t = 0, \qquad \mathrm{d}t\cdot\mathrm{d}B_t = 0, \qquad \mathrm{d}B_t\cdot\mathrm{d}B_t = \mathrm{d}t.$$

So for $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}B_t$: $\mathrm{d}\langle X\rangle_t = (\mathrm{d}X_t)^2 = (\mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}B_t)^2 = \sigma_t^2(\mathrm{d}B_t)^2 = \sigma_t^2\,\mathrm{d}t$, i.e. $\langle X\rangle_t = \int_0^t \sigma_s^2\,\mathrm{d}s$. The drift $\mu_t\,\mathrm{d}t$ contributes nothing (finite variation) — "drift is invisible" (Hack 22). Cross bracket: $\mathrm{d}\langle X, Y\rangle_t = \mathrm{d}X_t\,\mathrm{d}Y_t = \sigma_t^X\sigma_t^Y\,\mathrm{d}t$. Landmarks: $\mathrm{d}\langle B\rangle_t = \mathrm{d}t$, $\mathrm{d}\langle \sigma B\rangle_t = \sigma^2\,\mathrm{d}t$, $\mathrm{d}\langle S\rangle_t = \sigma^2 S_t^2\,\mathrm{d}t$ for GBM below.

M1 — The general Itô formula (integral form). Let X be an Itô process $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}B_t$, so $\mathrm{d}\langle X\rangle_t = \sigma_t^2\,\mathrm{d}t$. Then

$$f(X_t,t) = f(X_0,0) + \int_0^t \frac{\partial f}{\partial t}(X_s,s)\,\mathrm{d}s + \int_0^t \frac{\partial f}{\partial x}(X_s,s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \frac{\partial^2 f}{\partial x^2}(X_s,s)\,\mathrm{d}\langle X\rangle_s.$$

Substitute $\mathrm{d}X_s = \mu_s\,\mathrm{d}s + \sigma_s\,\mathrm{d}B_s$ and $\mathrm{d}\langle X\rangle_s = \sigma_s^2\,\mathrm{d}s$, then collect the $\mathrm{d}s$ and $\mathrm{d}B_s$ pieces:

$$f(X_t,t) = f(X_0,0) + \int_0^t \Big[\underbrace{\tfrac{\partial f}{\partial t}(X_s,s) + \mu_s\tfrac{\partial f}{\partial x}(X_s,s) + \tfrac{1}{2}\sigma_s^2\tfrac{\partial^2 f}{\partial x^2}(X_s,s)}_{\text{drift integrand}}\Big]\,\mathrm{d}s + \int_0^t \underbrace{\sigma_s\tfrac{\partial f}{\partial x}(X_s,s)}_{\text{martingale integrand}}\,\mathrm{d}B_s.$$

M1′ — Differential form (the $\mathrm{d}f(X_t,t)$ you asked for), everything evaluated at the running point (X_t,t):

$$\mathrm{d}f(X_t,t) = \frac{\partial f}{\partial t}(X_t,t)\,\mathrm{d}t + \frac{\partial f}{\partial x}(X_t,t)\,\mathrm{d}X_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(X_t,t)\,\mathrm{d}\langle X\rangle_t.$$

Brownian special case $X_t = B_t$ ($\mu_s = 0,\ \sigma_s = 1,\ \mathrm{d}\langle B\rangle_s = \mathrm{d}s$):

$$f(B_t,t) = f(B_0,0) + \int_0^t \Big[\tfrac{\partial f}{\partial t}(B_s,s) + \tfrac{1}{2}\tfrac{\partial^2 f}{\partial x^2}(B_s,s)\Big]\,\mathrm{d}s + \int_0^t \tfrac{\partial f}{\partial x}(B_s,s)\,\mathrm{d}B_s.$$

Recovers your hacks — each partial shown at its evaluation point:

- $f(x,t) = xt$: $\frac{\partial f}{\partial x}(B_s,s) = s$, $\frac{\partial f}{\partial t}(B_s,s) = B_s$, $\frac{\partial^2 f}{\partial x^2}(B_s,s) = 0 \Rightarrow tB_t = \int_0^t B_s\,\mathrm{d}s + \int_0^t s\,\mathrm{d}B_s$ (**Hack 17**).
- $f(x,t) = x^2$: $\frac{\partial f}{\partial x}(B_s,s) = 2B_s$, $\frac{\partial^2 f}{\partial x^2}(B_s,s) = 2$, $\frac{\partial f}{\partial t}(B_s,s) = 0 \Rightarrow B_t^2 = \int_0^t 2B_s\,\mathrm{d}B_s + \frac{1}{2}\int_0^t 2\,\mathrm{d}s = 2\int B\,\mathrm{d}B + t$ (**Hack 18**).
- $f(x,t) = e^{\theta x - \theta^2 t/2}$: $\frac{\partial f}{\partial x}(B_s,s) = \theta f(B_s,s)$, $\frac{\partial^2 f}{\partial x^2}(B_s,s) = \theta^2 f(B_s,s)$, $\frac{\partial f}{\partial t}(B_s,s) = -\frac{\theta^2}{2}f(B_s,s) \Rightarrow$ drift $= -\frac{\theta^2}{2}f + \frac{1}{2}\theta^2 f = 0$, pure $\mathrm{d}B$ (**Hack 4**).

M2 — Drift $= 0 \iff$ martingale (heat equation). From M1′, $f(B_t,t)$ is a (local) martingale iff its $\mathrm{d}t$-coefficient vanishes for all arguments:

$$\frac{\partial f}{\partial t}(x,t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(x,t) = 0 \qquad \text{for all } (x,t).$$

Dictionary check: $f = x^2 - t$: $-1 + \frac{1}{2}(2) = 0$; $f = x^3 - 3tx$: $-3x + \frac{1}{2}(6x) = 0$; $f = e^{t/2}\cos x$: $\frac{1}{2}e^{t/2}\cos x + \frac{1}{2}(-e^{t/2}\cos x) = 0$ (Hack 33's cos). All ✓.

M3 — Fully worked example: geometric Brownian motion. $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}B_t$; apply Itô with $f(x) = \log x$ (no explicit time).

Step 1 — derivatives: $\frac{\partial f}{\partial t} = 0$, $f'(x) = \frac{1}{x}$, $f''(x) = -\frac{1}{x^2}$; evaluated at S_t: $f'(S_t) = \frac{1}{S_t}$, $f''(S_t) = -\frac{1}{S_t^2}$.

Step 2 — the bracket (M0): here $\sigma_t = \sigma S_t$, so $\mathrm{d}\langle S\rangle_t = (\mathrm{d}S_t)^2 = (\sigma S_t\,\mathrm{d}B_t)^2 = \sigma^2 S_t^2\,\mathrm{d}t$.

Step 3 — plug into M1′ $\mathrm{d}f(S_t) = f'(S_t)\,\mathrm{d}S_t + \frac{1}{2}f''(S_t)\,\mathrm{d}\langle S\rangle_t$:

$$\mathrm{d}(\log S_t) = \frac{1}{S_t}(\mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}B_t) + \tfrac{1}{2}\Big(-\frac{1}{S_t^2}\Big)\sigma^2 S_t^2\,\mathrm{d}t = (\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B_t) - \tfrac{1}{2}\sigma^2\,\mathrm{d}t.$$

Step 4 — collect and integrate: $\mathrm{d}(\log S_t) = \big(\mu - \frac{1}{2}\sigma^2\big)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t \;\Rightarrow\; \boxed{S_t = S_0\,e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma B_t}}$.

The whole role of $\mathrm{d}\langle S\rangle_t = \sigma^2 S_t^2\,\mathrm{d}t$ is to feed the $\frac{1}{2}f''$ term, which is exactly the $-\frac{1}{2}\sigma^2$ Itô correction. Naive calculus would give only $\mu\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$; the $-\frac{1}{2}\sigma^2$ is $(\mathrm{d}B)^2 = \mathrm{d}t$ leaking in.

M4 — Itô product rule / general integration by parts. For two Itô processes X, Y:

$$\mathrm{d}(X_t Y_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}\langle X, Y\rangle_t, \qquad \langle X, Y\rangle_t = \int_0^t \sigma_s^X\,\sigma_s^Y\,\mathrm{d}s.$$

One factor finite-variation (e.g. deterministic t) $\Rightarrow$ $\langle X, Y\rangle = 0$ $\Rightarrow$ ordinary IBP (why **Hack 17** has no correction). Both $= B \Rightarrow \langle B, B\rangle_t = t \Rightarrow \mathrm{d}(B_t^2) = 2B_t\,\mathrm{d}B_t + \mathrm{d}t$ (**Hack 18**).

M5 — 2D Itô (the cross term). For $f = f(x, y, t)$ and Itô processes X, Y, with *all partials evaluated at* (X_t, Y_t, t):

$$\mathrm{d}f(X_t, Y_t, t) = \frac{\partial f}{\partial t}\,\mathrm{d}t + \frac{\partial f}{\partial x}\,\mathrm{d}X_t + \frac{\partial f}{\partial y}\,\mathrm{d}Y_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\,\mathrm{d}\langle X\rangle_t + \frac{1}{2}\frac{\partial^2 f}{\partial y^2}\,\mathrm{d}\langle Y\rangle_t + \frac{\partial^2 f}{\partial x\,\partial y}\,\mathrm{d}\langle X, Y\rangle_t.$$

Only new piece vs. 1D is $\frac{\partial^2 f}{\partial x\,\partial y}(X_t, Y_t, t)\,\mathrm{d}\langle X, Y\rangle_t$. Independent BMs $\Rightarrow$ $\mathrm{d}\langle X, Y\rangle = 0$; correlation $\rho \Rightarrow \mathrm{d}\langle W, B\rangle_t = \rho\,\mathrm{d}t$ — the engine behind **Hack 22**'s $W_t B_t - \rho t$ and **Hack 24**.

Calculus Toolkit — integrate, differentiate, solve (the moves used everywhere)

C1 — Differential $\longleftrightarrow$ integral form (the "integrate from 0 to t" move). An SDE in differential form is *shorthand* for an integral equation:

$$\mathrm{d}Y_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}B_t \quad\Longleftrightarrow\quad Y_t - Y_0 = \int_0^t a_s\,\mathrm{d}s + \int_0^t b_s\,\mathrm{d}B_s.$$

To integrate a differential: replace $\mathrm{d}Y_t \to Y_t - Y_0$, each $\mathrm{d}t \to \int_0^t(\cdots)\,\mathrm{d}s$, each $\mathrm{d}B_t \to \int_0^t(\cdots)\,\mathrm{d}B_s$, and rename the running time $t \to s$ inside. The lower limit 0 (with the constant Y_0) carries the initial condition; the upper limit t is the free variable. Limits must match the start point.

Example. $\mathrm{d}(\log S_t) = (\mu - \frac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t \Rightarrow \log S_t - \log S_0 = (\mu - \frac{1}{2}\sigma^2)t + \sigma B_t$ (used $\int_0^t \mathrm{d}s = t$, $\int_0^t \mathrm{d}B_s = B_t$).

C2 — Basic integrals you reuse. Deterministic: $\int_0^t c\,\mathrm{d}s = ct$, $\int_0^t s\,\mathrm{d}s = \frac{t^2}{2}$, $\int_0^t s^n\,\mathrm{d}s = \frac{t^{n+1}}{n+1}$, $\int_0^t e^{as}\,\mathrm{d}s = \frac{e^{at}-1}{a}$. Stochastic: $\int_0^t \mathrm{d}B_s = B_t$, $\int_0^t c\,\mathrm{d}B_s = cB_t$, and $\int_0^t h(s)\,\mathrm{d}B_s$ is a Wiener integral — Gaussian, mean 0, variance $\int_0^t h(s)^2\,\mathrm{d}s$ (Hack 1). Linearity holds for both: $\int_0^t(\alpha f + \beta g) = \alpha\int_0^t f + \beta\int_0^t g$.

C3 — Differentiating an integral (fundamental theorem + Leibniz, simple cases). The reverse of C1:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^t g(s)\,\mathrm{d}s = g(t) \quad \text{(FTC)}; \qquad \frac{\mathrm{d}}{\mathrm{d}t}\int_{a(t)}^{b(t)} g(s)\,\mathrm{d}s = g(b(t))\,b'(t) - g(a(t))\,a'(t).$$

Integrand also depending on t (fixed limits): $\dfrac{\mathrm{d}}{\mathrm{d}t}\displaystyle\int_\alpha^\beta g(t, s)\,\mathrm{d}s = \int_\alpha^\beta \frac{\partial g}{\partial t}(t, s)\,\mathrm{d}s$. Combined (full Leibniz): add the two boundary terms above to $\int_{a(t)}^{b(t)} \partial_t g(t, s)\,\mathrm{d}s$.

Trap. A stochastic integral $\int_0^t h\,\mathrm{d}B_s$ is *not* differentiable in t pathwise — never write $\frac{\mathrm{d}}{\mathrm{d}t}\int h\,\mathrm{d}B$. Handle $\mathrm{d}B$ terms only in differential form via Itô (M1′), not by ordinary differentiation.

C4 — Linear ODEs by integrating factor. For $y'(t) + p(t)\,y(t) = q(t)$, multiply by $\mu(t) = e^{\int p(t)\,\mathrm{d}t}$ so the left side collapses to a single derivative:

$$\big(\mu(t)y(t)\big)' = \mu(t)q(t) \;\Rightarrow\; y(t) = \frac{1}{\mu(t)}\Big[y_0\,\mu(0) + \int_0^t \mu(s)q(s)\,\mathrm{d}s\Big].$$

Constant coefficient $y' + ay = q(t)$: $\mu = e^{at}$, giving $y(t) = e^{-at}y_0 + \int_0^t e^{-a(t-s)}q(s)\,\mathrm{d}s$.

C5 — The SDE analog: integrating factor ⇒ Ornstein–Uhlenbeck. Same trick with an Itô term. For the linear SDE $\mathrm{d}X_t = (a - \theta X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$, multiply by $e^{\theta t}$. Since $e^{\theta t}$ is deterministic (finite variation, so *no* Itô correction — the Hack 17 phenomenon), the product rule gives $\mathrm{d}(e^{\theta t}X_t) = e^{\theta t}(\,\mathrm{d}X_t + \theta X_t\,\mathrm{d}t) = a\,e^{\theta t}\,\mathrm{d}t + \sigma e^{\theta t}\,\mathrm{d}B_t$. Integrate 0 to t (C1):

$$e^{\theta t}X_t - X_0 = \tfrac{a}{\theta}(e^{\theta t}-1) + \sigma\int_0^t e^{\theta s}\,\mathrm{d}B_s \;\Rightarrow\; \boxed{X_t = X_0e^{-\theta t} + \tfrac{a}{\theta}\big(1 - e^{-\theta t}\big) + \sigma\int_0^t e^{-\theta(t-s)}\,\mathrm{d}B_s}$$

The last term is a Wiener integral (C2): Gaussian, mean 0, variance $\sigma^2\int_0^t e^{-2\theta(t-s)}\,\mathrm{d}s = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t})$. This is the OU process, mean-reverting to a/θ. Same template solves any linear SDE $\mathrm{d}X_t = (\alpha_t + \beta_t X_t)\,\mathrm{d}t + (\gamma_t + \delta_t X_t)\,\mathrm{d}B_t$.

C6 — Vasicek model: the explicit Itô solve (not just the OU shortcut above). Model, in Vasicek's own letters (a = speed, b = long-run level, σ = vol):

$$\mathrm{d}r_t = a(b - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad r_0 \text{ given.}$$

Same SDE shape as C5 ($\theta \to a,\; a/\theta \to b$), but here is the derivation run explicitly through Itô's formula (M1′) rather than a bare product-rule trick. Take $f(x,t) = e^{at}x$: $\partial_t f = ae^{at}x$, $\partial_x f = e^{at}$, $\partial_{xx} f = 0$ (linear in x, so no second-order Itô term to worry about — verifying that IS the Itô step). By M1′,

$$\mathrm{d}\big(e^{at}r_t\big) = \partial_t f\;\mathrm{d}t + \partial_x f\;\mathrm{d}r_t = ae^{at}r_t\,\mathrm{d}t + e^{at}\big[a(b - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t\big].$$

The r_t terms cancel exactly ($ae^{at}r_t - ae^{at}r_t = 0$), leaving a differential with purely deterministic coefficients:

$$\mathrm{d}\big(e^{at}r_t\big) = ab\,e^{at}\,\mathrm{d}t + \sigma e^{at}\,\mathrm{d}W_t.$$

Integrate $0 \to t$ (C1, with C2's $\int_0^t e^{as}\,\mathrm{d}s = \frac{e^{at}-1}{a}$):

$$e^{at}r_t - r_0 = b(e^{at}-1) + \sigma\int_0^t e^{as}\,\mathrm{d}W_s \;\Longrightarrow\; \boxed{r_t = r_0e^{-at} + b\big(1 - e^{-at}\big) + \sigma\int_0^t e^{-a(t-s)}\,\mathrm{d}W_s}$$

This is the closed-form *path*, not just its mean: the last term is a Wiener integral (C2) with a deterministic integrand ⇒ Gaussian, mean 0. Reading off the distribution:

$$r_t \mid r_0 \;\sim\; N\Big(\underbrace{r_0e^{-at} + b(1 - e^{-at})}_{\text{mean}},\; \underbrace{\tfrac{\sigma^2}{2a}\big(1 - e^{-2at}\big)}_{\text{variance — Itô isometry, Hack 1}}\Big), \qquad t \to \infty:\; r_t \to N\Big(b, \tfrac{\sigma^2}{2a}\Big) \text{ (stationary).}$$

Trap. $f'' = 0$ here, so the "Itô correction" term is identically zero and the naive product rule already happens to be exact — that's why C5 could shortcut straight to it. That safety net breaks the moment the diffusion depends on r_t itself, which is exactly what happens next.

C7 — CIR model: mean only, by taking expectations (Itô gives no explicit solution here). Model:

$$\mathrm{d}r_t = a(b - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\;\mathrm{d}W_t.$$

Try the same integrating factor e^{at}: the drift piece cancels exactly as in C6, but the diffusion becomes $\sigma e^{at}\sqrt{r_t}\,\mathrm{d}W_t$ — the integrand $\sqrt{r_t}$ is *random*, so $\int_0^t e^{as}\sqrt{r_s}\,\mathrm{d}W_s$ has no closed form and **no explicit path formula for r_t exists**. The mean doesn't need the path though — only linearity of $\mathbb{E}$.
Step 1 (C1, integral form):

$$r_t = r_0 + \int_0^t a(b - r_s)\,\mathrm{d}s + \int_0^t \sigma\sqrt{r_s}\;\mathrm{d}W_s.$$

Step 2 (take $\mathbb{E}[\cdot \mid r_0]$): the Itô integral is a martingale started at 0, so its expectation is exactly 0 — this is precisely where the troublesome $\sqrt{r_s}$ disappears, without ever being evaluated. Writing $m(t) := \mathbb{E}[r_t \mid r_0]$ and sliding $\mathbb{E}$ through the $\mathrm{d}s$-integral (Fubini, Hack 20):

$$m(t) = r_0 + \int_0^t a\big(b - m(s)\big)\,\mathrm{d}s.$$

Step 3 (C3, differentiate to kill the integral):

$$m'(t) = a\big(b - m(t)\big), \qquad m(0) = r_0,$$

a deterministic linear ODE — identical in form to Vasicek's, because CIR and Vasicek share the *same drift*; only the diffusion differs, and diffusion never touches the mean (M0: it lives entirely inside $\mathrm{d}\langle r\rangle_t$, which $\mathbb{E}$ kills).
Step 4 (C4 integrating factor, or substitute $u = m - b \Rightarrow u' = -au$):

$$\boxed{\mathbb{E}[r_t \mid r_0] = r_0 e^{-at} + b\big(1 - e^{-at}\big)}$$

— *the same formula as Vasicek's mean*, since the derivation only ever used the drift.
Trap. This route gives the mean *only*. Unlike Vasicek, $r_t \geq 0$ always and the true law is non-central χ^2, not Gaussian. *Bonus* (same trick one level up, applying Itô to $\mathrm{d}(r_t^2)$ and taking $\mathbb{E}$ again):

$$\operatorname{var}(r_t \mid r_0) = r_0\,\tfrac{\sigma^2}{a}\big(e^{-at} - e^{-2at}\big) + b\,\tfrac{\sigma^2}{2a}\big(1 - e^{-at}\big)^2.$$

Hack 20: Fubini for $\mathrm{d}s$-integrals; the $t^3/3$ constants

$\mathbb{E}$ slides through ordinary integrals: $\mathbb{E}[\int_0^t B_s\,\mathrm{d}s] = \int_0^t \mathbb{E}[B_s]\,\mathrm{d}s = 0$; $\mathbb{E}[(\int_0^t B_s\,\mathrm{d}s)^2] = \int_0^t\int_0^t (r \wedge s)\,\mathrm{d}r\,\mathrm{d}s$. Constants:

$$\operatorname{var}\Big(\int_0^t B_s\,\mathrm{d}s\Big) = \tfrac{t^3}{3}, \quad \operatorname{cov}\Big(\int_0^s, \int_0^t\Big) = s^2\Big(\tfrac{t}{2} - \tfrac{s}{6}\Big), \quad \operatorname{var}\Big(\int_s^{s+t}\Big) = st^2 + \tfrac{t^3}{3}.$$

Last one $\neq t^3/3$ for $s > 0 \Rightarrow$ increments not stationary. Also $\int_0^t B_s\,\mathrm{d}s$ has finite variation and is nonconstant $\Rightarrow$ never a martingale.

Tier 4 — Products, brackets, covariances

Hack 21: Cross-isometry (polarization)

$$\mathbb{E}\Big[\int_0^t H\,\mathrm{d}B \cdot \int_0^t K\,\mathrm{d}B\Big] = \int_0^t \mathbb{E}[H_s K_s]\,\mathrm{d}s; \quad \text{different limits } s < t: \text{ integrate to the shorter one.}$$

Example. $\mathbb{E}[\int_0^t s\,\mathrm{d}B_s \cdot \int_0^t e^s\,\mathrm{d}B_s] = \int_0^t s e^s\,\mathrm{d}s = (t-1)e^t + 1$.
Example. $\mathbb{E}[B_t \int_0^t B_s\,\mathrm{d}B_s] = \int_0^t \mathbb{E}[B_s]\,\mathrm{d}s = 0$ (absorb B_t first, Hack 3).

Hack 22: Quadratic covariation; drift is invisible

$\big\langle \int H\,\mathrm{d}B, \int K\,\mathrm{d}B\big\rangle_t = \int_0^t H_s K_s\,\mathrm{d}s$, $\quad \langle X, A\rangle_t = 0$ if A has finite variation.
Companion: $X_t Y_t - \langle X, Y\rangle_t$ is a martingale (why $W_t B_t^{(2)} - \rho t$ works).
Example. $\langle B_\cdot + \mu\cdot, B\rangle_t = t + 0 = t$; $\quad \langle \int_0^\cdot s\,\mathrm{d}B_s, B\rangle_t = \int_0^t s\,\mathrm{d}s = \frac{t^2}{2}$.

Hack 23: Increment covariance = overlap length

$\operatorname{cov}(B_t - B_s, B_v - B_u) = \operatorname{length}\big((s,t] \cap (u,v]\big)$. Disjoint $\Rightarrow 0$; overlapping $\Rightarrow$ overlap length.
Example. $\operatorname{cov}(B_3 - B_1, B_4 - B_2) = |(2,3]| = 1$. One line, no algebra.

Hack 24: Correlated BMs — cross-terms die

For $W_t = \sqrt{1-\rho^2}\, B_t^{(1)} + \rho B_t^{(2)}$ (independent BMs): expand, use $\mathbb{E}[B_s^{(1)} B_t^{(2)}] = 0$.

$$\mathrm{cov}(W_s, B_t^{(2)}) = \rho(s \wedge t), \qquad \mathrm{corr}(W_t, B_t^{(2)}) = \rho.$$

General: $\sum_i \lambda_i B_t^{(i)}$ is a standard BM iff $\sum \lambda_i^2 = 1$ — checking the norm IS the question.

Tier 5 — Process identification & conditioning

Hack 25: The five invariances — BM in disguise

Each is again a standard BM: $-B_t$; $\lambda B_{t/\lambda^2}$ (e.g. $\frac{1}{\sqrt{c}} B_{ct}$); $B_{s+t} - B_s$; $tB_{1/t}$; $\sum \lambda_i B_t^{(i)}$ with $\sum \lambda_i^2 = 1$.
Verify when unsure: centred Gaussian $\Rightarrow$ just check cov $= s \wedge t$.
Example. $\frac{1}{2} B_{4t}$: cov $= \frac{1}{4}\min(4s, 4t) = s \wedge t$ ✓ BM.

Hack 26: "Is it Gaussian?" — two-second checklist

Linear in Gaussian inputs (sums, ds-integrals, deterministic time changes, Wiener integrals $\int h(s)\,\mathrm{d}B_s$) $\Rightarrow$ Gaussian ✓. Nonlinear ($B_t^2, e^{B_t}, |B_t|, \max_s B_s, \int B\,\mathrm{d}B$) $\Rightarrow$ not Gaussian ✗.
Example. $\int_0^t (t-s)\,\mathrm{d}B_s$ ✓. $\int_0^t B_s\,\mathrm{d}B_s = \frac{B_t^2 - t}{2}$: chi-square type ✗.

Hack 27: Gaussian conditioning — projection formula

Jointly Gaussian centred (X, Y): $\mathbb{E}[X \mid Y] = \dfrac{\mathrm{cov}(X,Y)}{\mathrm{var}(Y)} Y \Rightarrow \mathbb{E}[B_t \mid B_s] = B_s$, $\mathbb{E}[B_s \mid B_t] = \frac{s}{t} B_t$ $(s < t)$.
Trap. The backward one is the classic trap — the reflex answer B_t is wrong. Bonus: $\mathrm{var}(B_s \mid B_t) = \frac{s(t-s)}{t}$ (the bridge variance).
Example. $\mathbb{E}[B_2 \mid B_5] = \frac{2}{5} B_5$.

Hack 28: Gaussian-martingale transfer: replace t by $\Gamma(t,t)$

Everything true for BM with "t" holds for a continuous centred Gaussian martingale M with "$\Gamma(t,t)$": $\mathrm{var}(M_t) = \Gamma(t,t)$; $M_t^2 - \Gamma(t,t)$ mart.; $\langle M \rangle_t = \Gamma(t,t)$; $e^{\theta M_t - \frac{\theta^2}{2}\Gamma(t,t)}$ mart.
Example. $\mathbb{E}[M_t^2] = t^3$: for which θ is $e^{\theta M_t - 2t^3}$ a martingale? $\frac{\theta^2}{2} t^3 = 2t^3 \Rightarrow \theta = \pm 2$.

Tier 6 — Stopping, extremes, limits

Hack 29: Optional stopping / exit from an interval

For nice T: $\mathbb{E}[B_T] = 0$, $\mathbb{E}[B_T^2] = \mathbb{E}[T]$, $\mathbb{E}[e^{\theta B_T - \frac{\theta^2}{2} T}] = 1$. Master case T = exit of $[-a, b]$:

$$P(\text{hit } b \text{ before } -a) = \tfrac{a}{a+b}, \qquad \mathbb{E}[T] = ab.$$

Example. Exit of $[-1, 2]$: hits 2 first w.p. $\frac{1}{3}$; $\mathbb{E}[T] = 2$.
Trap. One-sided $T = \inf\{t : B_t = 1\}$: $\mathbb{E}[B_T] = 1 \neq 0$ — optional stopping fails ($\mathbb{E}[T] = \infty$).

Hack 30: Reflection principle — the running max for free

$M_t = \max_{s \le t} B_s$: $P(M_t \ge a) = 2P(B_t \ge a)$, $M_t \overset{d}{=} |B_t|$.
Example. $P(\max_{s \le 1} B_s \ge 1) = 2(1 - \Phi(1)) \approx 0.317$.
Example. $T_a = \inf\{t : B_t = a\}$: $P(T_a \le t) = 2P(B_t \ge a) \to 1$ — BM hits every level, yet $\mathbb{E}[T_a] = \infty$.

Hack 31: Long-run behavior — drift beats noise

$\frac{B_t}{t} \to 0$ a.s. $\Rightarrow B_t + \mu t \to \pm\infty$ (sign of μ), and $e^{\theta(B_t+\nu t)} \to 0$ if $\theta\nu < 0$, $\to \infty$ if $\theta\nu > 0$.
Trap. $S_t \to 0$ a.s. while $\mathbb{E}[S_t] \to \infty$ can both hold ($e^{B_t - 0.01t}$). Read whether the question asks a.s. or in expectation.

Hack 32: Doob's L^2 inequality — bound the sup

$\mathbb{E}[\sup_{s\le T} M_s^2] \le 4\,\mathbb{E}[M_T^2]$. Any bound-the-supremum question: "$4\times$ terminal second moment."
Example. $\mathbb{E}[\sup_{s\le 1}(B_{n+s} - B_n)^2] \le 4\mathbb{E}[(B_{n+1} - B_n)^2] = 4$.

Hack 33: Itô compensator — martingales from any f

$f(B_t) - \frac{1}{2}\int_0^t f''(B_s)\,\mathrm{d}s$ is a (local) martingale. Generates the whole dictionary of Hack 5.
Example. $f = \cos$: $\mathbb{E}[\cos B_t] = e^{-t/2}$, so $e^{t/2}\cos(B_t)$ is a martingale (constant mean 1 ✓).
See Master Forms M2: this is the time-independent slice of $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$.

Hack 34: MGF derivatives — mixed polynomial–exponential moments

$\mathbb{E}[e^{\theta B_t}] = e^{\theta^2 t/2}$, $\mathbb{E}[B_t e^{\theta B_t}] = \theta t\, e^{\theta^2 t/2}$, $\mathbb{E}[B_t^2 e^{\theta B_t}] = (t + \theta^2 t^2)e^{\theta^2 t/2}$.
(Differentiate the MGF in θ, or Stein: $\mathbb{E}[X f(X)] = \mathbb{E}[f'(X)]$.) **Example.** $\mathbb{E}[B_1 e^{B_1}] = e^{1/2}$.

Meta-hacks — answer elimination in 10 seconds

Hack 35: Sanity checks that kill wrong options

1. **Dimensions:** $B \sim \sqrt{t}$, so every term must carry a consistent total power of time. $\mathbb{E}[B_s^2 B_t^2]$ has degree 2: $s(t-s) + 3s^2$ ✓; a mixed-degree option like $s + 3s^2$ ✗.
2. **Endpoints:** set $s = t$ (variance ≥ 0), $t \to 0$ (must give 0), $t = 1$ for bridges (must give 0).
3. **Mean zero:** stochastic integrals and centred processes — discard any nonzero-mean option.
4. **Special values:** plug $\theta = 0, \rho = 0$: the formula must reduce to something known.
5. **Two-hack agreement:** if two independent hacks give the same number, it is the answer.

Every quiz question = one primary hack + one moment-table entry + 60 seconds of algebra. The skill being tested is recognizing which hack, fast.